Air University

AIR UNIVERSITY

Programming Fundamentals (Lab)

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Lab # 10

Lab 9.1 A while loop

Code

```
// This program displays a hot beverage menu and prompts the user to
// make a selection. A switch statement determines which item the user
// has chosen. A do-while loop repeats until the user selects item E
// from the menu.

// USAMA ARSHAD

#include <iostream>
#include <iomanip>
using namespace std;
int main()
{
    // Fill in the code to define an integer variable called number,
    // a floating point variable called cost,
    // and a character variable called beverage
    int number;
    float cost;
    char beverage;
    bool validBeverage;

    cout << fixed << showpoint << setprecision(2);

    do
    {
        cout << "\nA: Coffee $1.00" << endl;
```

```
        cout << "\nA: Coffee $1.00" << endl;
        cout << "B: Tea $ 0.75" << endl;
        cout << "C: Hot Chocolate $1.25" << endl;
        cout << "D: Cappuccino $2.50" << endl;
        cout << "\nEnter the beverage A,B,C, or D you desire "
        "\nOtherwise Enter E to EXIT" << endl;
        cin >> beverage;
        // Fill in the code to read in beverage
        switch(beverage)
        {
            case 'A': cout << "Coffee $1.00 selected " << endl ;
                break;
            case 'B': cout << "Tea $0.75 selected " << endl;
                break;
            case 'C': cout << "Hot Chocolate $1.25 selected " << endl;
                break;
            case 'D': cout << "Capuccino $2.50 selected " << endl;
                break;
            default: cout << "Invalid Selection " << endl;
        }
        if (beverage =='A'||beverage =='B'||beverage =='C'||beverage =='D')
        {
            cout << "How many cups would you like?" << endl;
            cin >> number ;

```

```
        switch (beverage)
        {
        case 'A': cost = number * 1.0;
            cout << "The total cost is $ " << cost << endl;
            break;
        case 'B': cost = number * 0.75;
            cout << "The total cost is $ " << cost << endl;
            break;
        case 'C': cost = number * 1.25;
            cout << "The total cost is $ " << cost << endl;
            break;
        case 'D': cost = number * 2.50;
            cout << "The total cost is $ " << cost << endl;
            break;
        default: cout << " Try again please" << endl;
        }
        }
    }
    while(beverage !='E' && beverage!='e');
    {
        cout << "Beverage is not Available " << endl;
    }
    return 0;
}
```

Output 1

```
A: Coffee $1.00
B: Tea $ 0.75
C: Hot Chocolate $1.25
D: Cappuccino $2.50

Enter the beverage A,B,C, or D you desire
Otherwise Enter E to EXIT
A
Coffee $1.00 selected
How many cups would you like?
1
The total cost is $ 1.00

A: Coffee $1.00
B: Tea $ 0.75
C: Hot Chocolate $1.25
D: Cappuccino $2.50

Enter the beverage A,B,C, or D you desire
Otherwise Enter E to EXIT
B
Tea $0.75 selected
How many cups would you like?
2
The total cost is $ 1.50

A: Coffee $1.00
B: Tea $ 0.75
C: Hot Chocolate $1.25
D: Cappuccino $2.50

Enter the beverage A,B,C, or D you desire
Otherwise Enter E to EXIT
C
Hot Chocolate $1.25 selected
How many cups would you like?
3
The total cost is $ 3.75
```

```
The total cost is $ 3.75

A: Coffee $1.00
B: Tea $ 0.75
C: Hot Chocolate $1.25
D: Cappuccino $2.50

Enter the beverage A,B,C, or D you desire
Otherwise Enter E to EXIT
D
Capuccino $2.50 selected
How many cups would you like?
4
The total cost is $ 10.00

A: Coffee $1.00
B: Tea $ 0.75
C: Hot Chocolate $1.25
D: Cappuccino $2.50

Enter the beverage A,B,C, or D you desire
Otherwise Enter E to EXIT
E
Invalid Selection
Beverage is not Available

--------------------------------
Process exited after 62.17 seconds with return value 0
Press any key to continue . . .
```

Output 2

```
D:\University\Programming Fundamentals (Lab)\Lab Tasks\Lab 10\Lab 9.1A while loop.exe

Enter the beverage A,B,C, or D you desire
Otherwise Enter E to EXIT
F
Invalid Selection

A: Coffee $1.00
B: Tea $ 0.75
C: Hot Chocolate $1.25
D: Cappuccino $2.50

Enter the beverage A,B,C, or D you desire
Otherwise Enter E to EXIT
G
Invalid Selection

A: Coffee $1.00
B: Tea $ 0.75
C: Hot Chocolate $1.25
D: Cappuccino $2.50

Enter the beverage A,B,C, or D you desire
Otherwise Enter E to EXIT
E
Invalid Selection
Beverage is not Available

--------------------------------
Process exited after 8.225 seconds with return value 0
Press any key to continue . . .
```

Lab 10 A – Task 1

Code

```cpp
#include <iostream>
using namespace std;
int main()
{
    //Declaring Variables
    int choice, i=1, coffee_sum=0, tea_sum=0, coke_sum=0, juice_sum=0 ;

    do
    {
        //Asking for favourite Beverages
        cout << "\nEnter favourite beverage for Person #" << i <<" :" << endl;
        cout << "Choose 1.Coffee 2.Tea 3.Coke 4.Orange Juice \nor Choose -1 to Exit" << endl;
        cin >> choice ;
        i++ ;

        if (choice == 1)
        {
            coffee_sum = coffee_sum + choice ;
        }
        if (choice == 2)
        {
            tea_sum = tea_sum + (choice - 1) ;
        }
        if (choice == 3)
        {
```

```cpp
        {
            coke_sum = coke_sum + (choice - 2) ;
        }
        if (choice == 4)
        {
            juice_sum = juice_sum + (choice - 3) ;
        }
    }
    while (choice!=-1);

    cout << "\nTotal Number of people Surveyed is " << i-2 << endl;
    cout << "\nBeverage \tNumber of Votes " << endl;
    cout << "**********************************" << endl;
    cout << "Coffee" << "\t\t\t" << coffee_sum << endl;
    cout << "Tea" << "\t\t\t" << tea_sum << endl;
    cout << "Coke" << "\t\t\t" << coke_sum << endl;
    cout << "Orange Juice" << "\t\t" << juice_sum << endl;

    return 0;
}
```

Output 1

```
Enter favourite beverage for Person #7 :
Choose 1.Coffee 2.Tea 3.Coke 4.Orange Juice
or Choose -1 to Exit
3

Enter favourite beverage for Person #8 :
Choose 1.Coffee 2.Tea 3.Coke 4.Orange Juice
or Choose -1 to Exit
3

Enter favourite beverage for Person #9 :
Choose 1.Coffee 2.Tea 3.Coke 4.Orange Juice
or Choose -1 to Exit
-1

Total Number of people Surveyed is 8

Beverage        Number of Votes
*********************************
Coffee                  2
Tea                     2
Coke                    2
Orange Juice            2

--------------------------------
Process exited after 24.56 seconds with return value 0
Press any key to continue . . .
```

Output 2

```
D:\University\Programming Fundamentals (Lab)\Lab Tasks\Lab 10\Lab10A-Task1.exe

Enter favourite beverage for Person #1 :
Choose 1.Coffee 2.Tea 3.Coke 4.Orange Juice
or Choose -1 to Exit
1

Enter favourite beverage for Person #2 :
Choose 1.Coffee 2.Tea 3.Coke 4.Orange Juice
or Choose -1 to Exit
1

Enter favourite beverage for Person #3 :
Choose 1.Coffee 2.Tea 3.Coke 4.Orange Juice
or Choose -1 to Exit
4

Enter favourite beverage for Person #4 :
Choose 1.Coffee 2.Tea 3.Coke 4.Orange Juice
or Choose -1 to Exit
4

Enter favourite beverage for Person #5 :
Choose 1.Coffee 2.Tea 3.Coke 4.Orange Juice
or Choose -1 to Exit
2

Enter favourite beverage for Person #6 :
Choose 1.Coffee 2.Tea 3.Coke 4.Orange Juice
or Choose -1 to Exit
2
```

Lab 10 B – Task 1

Code

```cpp
// This program has the user input a number n and then finds the
// mean of the first n positive integers
// USAMA ARSHAD
#include <iostream>
using namespace std;

int main()
{
    int value; // value is some positive number n
    int total = 0; // total holds the sum of the first n positive numbers
     float mean=0.0;
     int _1stNo, _2ndNo;

    cout << "Enter Two positive Integers to find their mean" << endl;
    cout << "\nEnter first positive integer" << endl;
    cin >> _1stNo;
    cout << "Enter second positive integer" << endl;
    cin>>_2ndNo;

    //Handling 0
    if ( _1stNo > 0 && _2ndNo > 0)
    {
        //for Loop
        for (int number = _1stNo; number <= _2ndNo; number++)
        {
```

```cpp
    {
        total = total + number;
    }
        mean = static_cast<float>(total) / ((_2ndNo-_1stNo) +1 ); // note the use of the typecast

    cout << "The mean  of the positive integers " << _1stNo <<" and "<<_2ndNo
    << "  is " << mean << endl;
}
else
    cout << "Invalid input - integer must be positive" << endl;

return 0;
}
```

Output

```
D:\University\Programming Fundamentals (Lab)\Lab Tasks\Lab 10\Lab10B-Task1.exe
Enter Two positive Integers to find their mean

Enter first positive integer
3
Enter second positive integer
9
The mean  of the positive integers 3 and 9  is 6

--------------------------------
Process exited after 3.432 seconds with return value 0
Press any key to continue . . .
```

Lab 10 B – Task 2

Code

```cpp
#include <iostream>
using namespace std;
int main ()
{
    //declaring variables
    int speed, time;
    float distance;

    //asking user for speed and time and performing input validation
    cout << "What is the speed of vehicle in mph ?" << endl;
    cin >> speed ;
    while (speed <= 0)
    {
        cout << "Kindly Enter speed atleast 1 mph." << endl;
        cin >> speed ;
    }
    cout << "How many hours the vehicle has traveled ?" << endl;
    cin >> time ;
    while (time < 1)
    {
        cout << "Kindly Enter Time atleast 1 hour." << endl;
        cin >> time;
```

```cpp
        cout << "Kindly Enter Time atleast 1 hour." << endl;
        cin >> time;
    }

    //calculating total distance
    distance = speed * time ;
    cout << "\nTotal Distance traveled is " << distance << " miles" << endl;

    //printing pattern
    cout << "\nHours\t Distance Traveled" << endl;
    cout << "-------------------------" << endl;

        //calculating distance covered per each hour
        for (int i=1 ; i<=time ; i++)
        {
            distance = speed * i ;

            cout << i << "\t\t" << distance << endl;
        }
    return 0;
}
```

Output

```
D:\University\Programming Fundamentals (Lab)\Lab Tasks\Lab 10\Lab10B-Task2.exe
What is the speed of vehicle in mph ?
0
Kindly Enter speed atleast 1 mph.
-3
Kindly Enter speed atleast 1 mph.
40
How many hours the vehicle has traveled ?
0
Kindly Enter Time atleast 1 hour.
-5
Kindly Enter Time atleast 1 hour.
3

Total Distance traveled is 120 miles

Hours    Distance Traveled
-------------------------
1               40
2               80
3               120

--------------------------------
Process exited after 16.06 seconds with return value 0
Press any key to continue . . .
```

Lab 10.1 C nested loop

Code

```cpp
// This program finds the average time spent programming by a student
// each day over a three day period.
// USAMA ARSHAD

#include <iostream>
using namespace std;

int main()
{
    int num_students;
    float num_hours, average, total_avg=0;
    int student,day = 0; // these are the counters for the loops

    cout << "This program will find the average number of hours a day"
    << " that a student spent programming over a long weekend\n";

    cout << "\nHow many students are there ?" << endl;
    cin >> num_students ;

    for(student = 1; student <= num_students; student++)
    {
        float total=0;
        for(day = 1; day <= 3; day++)
        {
            cout << "\nEnter number of hours worked by student "
                << student <<" on day " << day << "." << endl;
            cin >> num_hours;

            total += num_hours ;
        }
        average = total / 3;
        total_avg += average ;
        cout << "\nThe average number of hours per day spent programming by "
            << "student " << student << " is " << average
            << endl;
    }
    cout << "\nThe average number of hours programmed by students is " << total_avg << endl;
    return 0;
}
```

Output

```
D:\University\Programming Fundamentals (Lab)\Lab Tasks\Lab 10\Lab10.1C nested loop.exe
How many students are there ?
2

Enter number of hours worked by student 1 on day 1.
1

Enter number of hours worked by student 1 on day 2.
2

Enter number of hours worked by student 1 on day 3.
3

The average number of hours per day spent programming by student 1 is 2

Enter number of hours worked by student 2 on day 1.
2

Enter number of hours worked by student 2 on day 2.
3

Enter number of hours worked by student 2 on day 3.
4

The average number of hours per day spent programming by student 2 is 3

The average number of hours programmed by students is 5

--------------------------------
Process exited after 7.362 seconds with return value 0
Press any key to continue . . .
```

Lab 10 C – Task 1

Code

```
#include <iostream>
using namespace std;
int main()
{
    // declaring variables
    int num;

    // taking input from user
    cout << "Enter number you want to see tables. " ;
    cin >> num;

    // checking validation
    while (num<1)
    {
        cout << "Kindly enter any number greater than 0. " << endl;
    }

    // printing tables
    for (int i=1 ; i <= num ; i++)
    {
        for (int j=1 ; j<=10 ; j++)
        {
        cout << "\t" << i*j << " " ;
        }
        cout << endl;
    }
    return 0;
}
```

Output

```
D:\University\Programming Fundamentals (Lab)\Lab Tasks\Lab 10\Lab10C-Task1.exe
Enter number you want to see tables. 12
        1       2       3       4       5       6       7       8       9       10
        2       4       6       8       10      12      14      16      18      20
        3       6       9       12      15      18      21      24      27      30
        4       8       12      16      20      24      28      32      36      40
        5       10      15      20      25      30      35      40      45      50
        6       12      18      24      30      36      42      48      54      60
        7       14      21      28      35      42      49      56      63      70
        8       16      24      32      40      48      56      64      72      80
        9       18      27      36      45      54      63      72      81      90
        10      20      30      40      50      60      70      80      90      100
        11      22      33      44      55      66      77      88      99      110
        12      24      36      48      60      72      84      96      108     120

--------------------------------
Process exited after 2.936 seconds with return value 0
Press any key to continue . . .
```

Lab 10 C – Task 2

Code

```
#include <iostream>
using namespace std;
int main ()
{
    // declaring variables
    int minutes, seconds ;

    //taking inputs
    cout << "Enter number of minutes " << endl;
    cin >> minutes ;

    //type validation
    while ( minutes<0 )
    {
        cout << "Kindly enter atleast 0 minutes " << endl;
        cin >> minutes ;
    }
    cout << "Enter number of seconds " << endl;
    cin >> seconds ;
    while ( seconds<1 )
    {
        cout << "Kindly enter seconds between (0 - 59) " << endl;
        cin >> seconds ;
    }

    //loop for timer
    for (int i=minutes ; i>=0 ; i-- )
    {
        for (int j=seconds ; j>=0 ; j-- )
        {
            cout << i << " Minutes " << j << " Seconds " << endl;
            cout.flush();
        }
        seconds = 59;
    }
    return 0;
}
```

Output

```
D:\University\Programming Fundamentals (Lab)\Lab Tasks\Lab 10\Lab10C-Task2.exe
Enter number of minutes
1
Enter number of seconds
12
1 Minutes 12 Seconds
1 Minutes 11 Seconds
1 Minutes 10 Seconds
1 Minutes 9 Seconds
1 Minutes 8 Seconds
1 Minutes 7 Seconds
1 Minutes 6 Seconds
1 Minutes 5 Seconds
1 Minutes 4 Seconds
1 Minutes 3 Seconds
1 Minutes 2 Seconds
1 Minutes 1 Seconds
1 Minutes 0 Seconds
0 Minutes 59 Seconds
0 Minutes 58 Seconds
0 Minutes 57 Seconds
0 Minutes 56 Seconds
0 Minutes 55 Seconds
0 Minutes 54 Seconds
0 Minutes 53 Seconds
0 Minutes 52 Seconds
0 Minutes 51 Seconds
0 Minutes 50 Seconds
0 Minutes 49 Seconds
0 Minutes 48 Seconds
0 Minutes 47 Seconds
0 Minutes 46 Seconds
0 Minutes 45 Seconds
```

```
D:\University\Programming Fundamentals (Lab)\Lab Tasks\Lab 10\Lab10C-Task2.exe
0 Minutes 47 Seconds
0 Minutes 46 Seconds
0 Minutes 45 Seconds
0 Minutes 44 Seconds
0 Minutes 43 Seconds
0 Minutes 42 Seconds
0 Minutes 41 Seconds
0 Minutes 40 Seconds
0 Minutes 39 Seconds
0 Minutes 38 Seconds
0 Minutes 37 Seconds
0 Minutes 36 Seconds
0 Minutes 35 Seconds
0 Minutes 34 Seconds
0 Minutes 33 Seconds
0 Minutes 32 Seconds
0 Minutes 31 Seconds
0 Minutes 30 Seconds
0 Minutes 29 Seconds
0 Minutes 28 Seconds
0 Minutes 27 Seconds
0 Minutes 26 Seconds
0 Minutes 25 Seconds
0 Minutes 24 Seconds
0 Minutes 23 Seconds
0 Minutes 22 Seconds
0 Minutes 21 Seconds
0 Minutes 20 Seconds
0 Minutes 19 Seconds
0 Minutes 18 Seconds
0 Minutes 17 Seconds
0 Minutes 16 Seconds
0 Minutes 15 Seconds
0 Minutes 14 Seconds
0 Minutes 13 Seconds
0 Minutes 12 Seconds
0 Minutes 11 Seconds
0 Minutes 10 Seconds
0 Minutes 9 Seconds
0 Minutes 8 Seconds
0 Minutes 7 Seconds
0 Minutes 6 Seconds
0 Minutes 5 Seconds
0 Minutes 4 Seconds
0 Minutes 3 Seconds
0 Minutes 2 Seconds
0 Minutes 1 Seconds
0 Minutes 0 Seconds

--------------------------------
Process exited after 3.429 seconds with return value 0
Press any key to continue . . .
```

Lab 10 C – Task 3

Code

```cpp
#include <iostream>
using namespace std;
int main()
{
    //Upperside
    for (int i=1 ; i<=5 ; i++)
    {
        for (int j=i ; j<5 ; j++)
        {
            cout << " " ;
        }
        for (int j=1 ; j<=(2*i-1) ; j++)
        {
            cout << "*" ;
        }
        cout << endl;
    }
    //Lowerside
    for (int i=4 ; i>=1 ; i--)
    {
        for (int j=5 ; j>i ; j--)
        {
        cout << " " ;
        }
        for (int j=1 ; j<=(2*i-1) ; j++)
        {
            cout <<"*" ;
        }
        cout << endl;
    }
    return 0;
}
```

Output

```
D:\University\Programming Fundamentals (Theory)\Activity\Nested Loop (Pa
    *
   ***
  *****
 *******
*********
 *******
  *****
   ***
    *

--------------------------------
Process exited after 0.249 seconds with return value 0
Press any key to continue . . .
```